

Levi Harris

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RESEARCH INTERESTS

Precipitation estimation, computer vision

EDUCATION

Northeastern University

Ph.D. Interdisciplinary Engineering – *Advisor: Auroop Ganguly*

Boston, MA
Fall 2026-Present

- Primary department: Civil & Environmental Engineering
- *Alsaif Doctoral Fellow*

University of North Carolina at Chapel Hill

M.S. Computer Science – *Advisor: Tianlong Chen*

B.S. Computer Science

Chapel Hill, NC
Fall 2025
Spring 2024

- Minor: Data Science

EXPERIENCE

University of Texas at Austin

Research Scientist I, *Jackson School of Geosciences, Advisor: Dev Niyogi*

Austin, TX
Summer 2026

University of North Carolina at Chapel Hill

GRA, *Department of Computer Science, Advisor: Tianlong Chen*

URA, *Department of Computer Science, Advisor: Gedas Bertasius*

Chapel Hill, NC
Fall 2024-Winter 2025
Spring 2023-Fall 2024

The National Weather Service

Summer volunteer, *Mentor: Stanley Czyzyk*

Las Vegas, NV
Summer 2025

SELECTED PUBLICATIONS

- 1 Levi Harris, Md Jayed Hossain, Mufan Qiu, Ruichen Zhang, Pingchuan Ma, Tianlong Chen, Jiaqi Gu, Seth Ariel Tongay, Umberto Celano, “SparseC-AFM: a deep learning method for fast and accurate characterization of MoS₂ with C-AFM,” *SPIE Conference on Low-Dimensional Materials and Devices; paper 13582-35*, 2025.

PRE-PRINTS

- 1 Oscar Ovanger, Levi Harris, Timothy H. Keitt, “Adaptive Evidence Weighting for Audio-Spatiotemporal Fusion”, *arXiv preprint arXiv:2602.03817*, 2026.
- 2 Levi Harris, Yagmur Derin, Pierre Kirstetter, “Learning probabilistic models of radar rainfall error”, 2026. *In Preparation for the Journal of Hydrometeorology*
- 3 Levi Harris, Yagmur Derin, Pierre Kirstetter, “Sub-Cloud Evaporation Dominates Rainfall Estimates in the Mojave Desert”, 2026. *In Preparation for Geophysical Research Letters*
- 4 Levi Harris, Tianlong Chen, “A Space-Time Transformer for Precipitation Nowcasting”, **1st Place NeurIPS 2025 Weather4Cast Challenge**, *arXiv preprint arXiv:2511.11090*, 2025.
- 5 Levi Harris, “A Simple and Effective Temporal Grounding Pipeline for Basketball Broadcast Footage”, *arXiv preprint arXiv:2411.00862*, 2024.

TALKS/PANEL DISCUSSIONS

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| 1. “ SaTformer ” [Oral]
<i>NeurIPS 2025 Weather4Cast Challenge</i> | 6 December 2025
San Diego, CA |
| 2. “ SparseC-AFM ” [Author]
<i>SPIE Conference on Low-Dimensional Materials and Devices 2025</i> | 17 September 2025
San Diego, CA |
| 3. “ Decoding Graduate Research at UNC ” [Panelist]
<i>UNC CS for Social Good</i> | 14 October 2025
Chapel Hill, NC |
| 4. “ Life as an undergraduate researcher ”
<i>UNC AI Club Symposium</i> | 29 November 2023
Chapel Hill, NC |

AWARDS

Northeastern University

- Alsaif Doctoral Fellowship 2026

University of North Carolina at Chapel Hill

- Department Tuition Award 2024, 2025
- Undergraduate Research Stipend 2023, 2024
- Dean’s List 2022, 2023, 2024

TEACHING

DATA 120: Ethics of Artificial Intelligence

<i>Teaching Assistant — Instructor: Santiago Olivella</i>	<i>Fall 2025</i>
<i>Teaching Assistant — Instructor: Justin Sola</i>	<i>Spring 2025</i>
<i>Teaching Assistant — Instructor: Neil Gaikwad</i>	<i>Fall 2024</i>

MENTORING

- **Asyat Sow** (UNC Chancellor’s Scholar) *Summer 2025*
Undergraduate · Project Mentor
“*Failure Modes of Real-Time Speech-to-Speech Translation Systems on French Regional Dialects*”
- **Wonjae Oh** *Fall 2024*
Undergraduate · Project Mentor
“*A Day-One Jailbreak for GPT-4V using ASCII Ciphers*”

SKILLS

Programming Languages	Python (fluent), Java
Libraries	PyTorch, TensorFlow, Angular, Spring Boot, UIKit
Databases	Oracle, Firebase
Tools	High-Performance Computing (HPC), Git, Docker, Vertex AI
SPM	Gwyddion, MATLAB, ImageJ

SELECTED COURSES

Multivariate Calculus · Introduction to Probability · Linear Algebra · Introduction to Machine Learning · Computational Imaging · Video Recognition with Transformers · Computer Vision of our 3D World · Efficient Machine Learning · Transfer Learning

REFERENCES

Pierre Kirstetter

Professor, School of Meteorology, University of Oklahoma

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Website

Tianlong Chen

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Website

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Neil Gaikwad

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